

LOHIT ASHWA VASWANI

AI Engineer · Data Scientist · LLM Systems Builder

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PROFESSIONAL SUMMARY

AI Engineer with an MSc Data Science Distinction from the University of Surrey and over two years of experience taking AI systems from prototype to production. Currently building two live exam preparation platforms with paying users: Sparkmed (LLM-graded medical exam preparation for MRCP, MRCPCH, and FRCS) and Markaroo (LLM-marked A-Level Economics tutoring against AQA mark schemes). Day-to-day work spans LLM orchestration and prompt engineering, asynchronous backend architecture, evaluation, and shipping the result. Earlier delivered an MSc thesis with the UK Health Security Agency benchmarking frontier LLMs on WHO Joint External Evaluation reports (currently being extended for publication in PLOS Journal), and a Llama-2 trading assistant for Trade for Sight that reached 70% adoption among active traders within three months. Comfortable owning a problem end-to-end: framing it, building it, evaluating it, and keeping it running.

TECHNICAL SKILLS

Programming: Python, SQL, NoSQL, JavaScript, C/C++

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face Transformers

Agentic AI and LLMs: LangChain, LlamaIndex, Prompt Engineering, Context Engineering, LLM Orchestration, MCP, A2A Protocol, QLoRA, PeFT

Databases & Vector Search: PostgreSQL, Oracle, MongoDB, Redis, FAISS, Pinecone

Cloud and DevOps: GCP (Vertex AI), AWS (EC2, SageMaker, Lambda), Docker, Docker Compose, Nginx, MLflow, Jenkins, GitHub Actions, Git, CI/CD

APIs and Backend: FastAPI, Flask, Django, Celery

Data Engineering: Apache Kafka, Databricks, Spark, ETL Pipelines, Tableau, Power BI, Selenium

PROFESSIONAL EXPERIENCE

AI Engineer | *Sparkmed and Markaroo, London, UK (Remote)*

May 2026 – Present

Building production AI-powered exam preparation platforms for high-stakes assessments; both products are deployed with active users including paying customers.

Sparkmed — AI-First Medical Education Platform (Production SaaS)

- **LLM grading pipeline:** Designed a Google Gemini-backed grading engine that evaluates free-text medical answers across Viva, SBA, OSCE, and SJT exam formats, returning structured clinical feedback with correctness scoring and explanations tailored to each question type. Engineered prompts to enforce examiner-style reasoning and consistent output schemas for downstream UI rendering.
- **Asynchronous inference architecture:** Built a Celery and Redis task queue that decouples LLM grading latency from the HTTP request cycle, allowing the platform to handle multi-second model calls on complex clinical questions without blocking the user interface, with retry, timeout, and failure-handling logic for reliability.
- **Personalised coaching engine:** Engineered a per-candidate readiness model that tracks blueprint coverage across clinical specialties, predicts MRCP, MRCPCH, and FRCS pass likelihood after 480+ attempted questions, and adapts session content based on detected study state (flow, challenge, recovery) to keep candidates progressing without burning out or coasting.
- **Rules-based intervention layer:** Implemented a deterministic intervention engine that evaluates post-session candidate state and triggers targeted AI-generated coaching messages only when criteria are met, with full audit logging of every clinical coaching decision to support traceability and safe deployment in a regulated domain.
- **Sparkbot conversational assistant:** Shipped a Gemini-backed Q&A assistant with medical-domain prompt engineering, persistent session context, and per-user rate limiting, enabling students to ask follow-up clinical questions inline during exam practice without leaving the platform.
- **Automated content ingestion:** Developed a PDF-upload, OCR (pytesseract), and LLM-based question-extraction pipeline that converts medical textbooks into structured, exam-ready clinical questions, significantly reducing the manual editorial effort required to grow the question bank.
- **Production deployment:** Containerised the platform with Docker Compose orchestrating Flask, Celery, Redis, PostgreSQL, and Nginx, with Jenkins CI/CD covering build, database migrations, and zero-downtime restarts; the system is live and serving paying customers.

Markaroo — AQA A-Level Economics AI Tutoring Platform

- **Rubric-aware LLM marking system:** Built an automated marking engine using Moonshot's Kimi K2.6 that grades student answers against official AQA mark schemes, with distinct prompting strategies for point-based rubrics (condition, accept, reject logic) and level-based rubrics (band descriptors with indicative content), replicating the way human examiners apply each rubric type.
- **Self-healing OCR-to-JSON pipeline:** Designed a two-stage extraction pipeline combining Mistral OCR for layout-aware text extraction with LLM-driven structured extraction into a target JSON schema. Validation errors are re-injected into the next prompt as corrective feedback, mitigating hallucination and schema drift across retries; achieved first-pass extraction success on 7 of 8 mark scheme documents covering 2017 to 2024.
- **Mark scheme normalisation:** Consolidated eight years of heterogeneous AQA mark schemes into a single consistent JSON schema, with Cloudinary-hosted diagram assets extracted directly from source PDFs, producing a clean reference corpus that powers the marking system and supports future model evaluation.
- **API and resilience layer:** Exposed the marking engine through a Flask API with JWT authentication, per-user rate limiting, and graceful fallback behaviour on LLM quota or API errors, so the platform stays responsive even when the upstream provider is degraded.
- **Deployment:** Containerised the full stack with Docker and Nginx and automated build, test, and release through GitHub Actions, enabling fast and repeatable deployments.

Data Scientist Intern | *CreatorOS, London, UK (Remote)*

Sep 2025 – Dec 2025

- Designed and deployed a two-stage ML recommendation system to solve CreatorOS's core product problem of unreliable brand-to-creator matching. Conducted feature engineering and statistical analysis on behavioural signals, then developed and validated a pruned RandomForest Classifier for creator segmentation and an XGBoost regression model for campaign-level ranking. Deployed via FastAPI and Docker on GCP Vertex AI at under 250ms latency. Applied MistralOCR to automate structured data extraction from unstructured client briefs directly into PostgreSQL. Production outcomes: 12% improvement in match precision (Precision@10), 10% improvement in ranking quality (NDCG@10), and 63% reduction in manual operational overhead.

Data Engineer, Client Contract | *via Facebook, Karachi (Remote)***Mar 2025 – May 2025**

- Led the Change of Value (COV) processing project for Niagara Bottling's IoT device manufacturing systems using Databricks Workflows and Spark SQL, building real-time streaming pipelines into the bronze layer. Designed fault-tolerant logic to handle power and ISP outages cutting streams mid-flow, out-of-sequence data arrival from streaming delays, and partial data windows from incomplete sensor readings, ensuring reliable end-to-end data processing regardless of device disruption.
- On a second engagement, the client needed an AI system to generate personalised travel itineraries from a large, inconsistent multi-source dataset. Raw data from all sources was first unified into a structured JSON format using Spark ETL pipelines in Databricks, then indexed into a FAISS vector database for semantic retrieval and to reduce hallucination of LLM. A three-agent LLM system was then built using LangChain and the DBRX model: a planner agent to decompose the user's travel request, a retriever agent to query the FAISS index, and an executor agent to assemble the final itinerary, with prompt engineering applied at every stage to keep responses grounded and hallucination-free.

Research Engineer (Promoted from Intern) | *Retrocausal, Redmond, USA (Remote)***Feb 2024 – Jan 2025****Ergonomics Report Generator**

- Designed and built an automated Excel reporting tool computing ergonomic risk scores including REBA, RULA, NIOSH, SNOOK, and Hand Strain from sensor data, fully integrated with the company's Django backend via POST APIs, eliminating manual spreadsheet preparation. Refactored the tool into a standalone FastAPI microservice containerised with Docker for independent deployment, horizontal scaling, and simplified CI/CD integration. Reduced report generation time by 44%, directly improving client-facing turnaround.

Model Optimisation (OpenVINO)

- Converted the production YOLOv5 pose-estimation model from PyTorch to OpenVINO IR format and applied INT8 post-training quantisation for CPU-based edge deployment without requiring GPU hardware. Achieved 2x faster inference throughput and reduced compute costs by 36% with negligible drop in keypoint detection accuracy.

Industrial Video Captioning System

- Conducted a structured comparative study fine-tuning two video-language architectures on internal assembly line footage: Timesformer+GPT-2 with temporal attention encoding and ViT+GPT-2 with frame-level encoding, analysing trade-offs between temporal modelling depth and inference efficiency to guide production model selection. The deployed captioning system reduced assembly errors by 25% and cut rework costs by 30%.

ML and Product Development Engineer Intern | *Transaction Processing System, Karachi***Jul 2023 – Aug 2023**

- Built a prototype ATM withdrawal recommendation model using XGBoost with temporal and behavioural features including transaction recency, frequency, and time-of-day, reducing RMSE by 8% over baseline heuristics. Designed the end-to-end ML pipeline covering data preprocessing, feature engineering, model training, and validation on 300,000+ ATM transactional records sourced via Oracle SQL extraction workflows that cut data preparation time by 20% and improved query efficiency for downstream analysis.
- Containerised the model as a Docker image and deployed it on the bank's internal server, exposing prediction endpoints to internal cash management workflows. Ran a real pilot across 200+ end users, showing 5 to 6% potential reduction in cash stockout events. Instrumented the service with structured prediction logging and configured a scheduled retraining job so the model could adapt to drifting withdrawal patterns over time, working with product and engineering teams to align outputs with operational requirements and contributing to a production proof-of-concept for improving ATM cash management decisions.

Teaching and Academic Support | *NUCES FAST University, Karachi***Aug 2022 – May 2024**

- **Student Lab Assistant, OOP (Aug 2022 – Dec 2022):** Mentored students on object-oriented principles including inheritance, polymorphism, and abstraction, guided debugging sessions, and reinforced software engineering best practices across weekly lab sessions.
- **Teaching Assistant, AI Programming (Aug 2023 – Dec 2023):** Delivered structured sessions on search algorithms, algorithmic game theory, and knowledge representation and reasoning; graded assignments and guided students in implementing AI models from scratch.
- **Teaching Assistant, Recommendation Systems (Jan 2024 – May 2024):** Guided students through collaborative filtering and neural recommender models; facilitated seminars on model evaluation, hyperparameter optimisation, and ethical considerations in recommender system design.

SELECTED PROJECTS**Automating WHO JEE Report Evaluation Using LLMs** | *MSc Thesis, Distinction* | **UK Health Security Agency****Jun – Sep 2025***Mistral OCR | Gemini 2.5 Flash | Qwen-3 | Magistral-Medium | BERTScore | QWK | Python*

- Addressed the problem of inconsistent and labour-intensive scoring in WHO Joint External Evaluation reports by designing a fully automated LLM evaluation pipeline capable of replicating human annotation at scale. Used MistralOCR to parse and structure text from 100+ page PDFs into indicator-level JSON, then constructed a labelled dataset with rubric-based scores across Evidence Quality and Actionability dimensions.
- Benchmarked Gemini 2.5 Flash, Qwen-3, and Magistral-Medium against human annotations using Quadratic Weighted Kappa and BERTScore, demonstrating that frontier LLMs can produce consistent, reproducible evaluation of complex WHO reporting frameworks. Presented as a Proof of Concept at the UK Health Security Agency and currently extending for peer-reviewed publication in PLOS Journal.

Optimisation Methods for Mixed Discrete–Continuous LoRA Hyperparameters**Nov 2025***Python | PyTorch | HuggingFace Transformers | Metaheuristic Algorithms*

- Developed a hyperparameter optimisation framework for LoRA-based fine-tuning of DistilBERT on a multi-class emotion classification task under strict GPU constraints. Implemented and compared five optimisation strategies: Random Search, Aquila Optimizer, Discrete Particle Swarm Optimisation with Genetic Algorithm, Blindfolded Spiderman Optimisation, and Multi-Objective Evolutionary Algorithm based on Decomposition with Differential Evolution, with a consistent experimental setup across all methods. Achieved 94% validation accuracy using BSO, outperforming population-based methods while converging faster.

Fraud Detection Benchmarking: Ensemble Aggregation vs Advanced Models under Class Imbalance**Dec 2025***Python | Scikit-learn | Ensemble Learning | Advanced Models | Aggregation Functions*

- Built a fraud detection system on 300,000+ transactions with severe class imbalance of approximately 2% fraud, applying SMOTE-Tomek for imbalance handling. Developed ensemble models combining Logistic Regression, KNN, Random Forest, and Naive Bayes with aggregation strategies including entropy-based, weighted average, threshold-aware, max probability, and trimmed mean. Benchmarked advanced models including XGBoost, Linear SVM, LDA, RIPPER, and Autoencoder+MLP using ROC-AUC, PR-AUC, precision, recall, and F1-score. Achieved 91% recall and reduced false positives from 10,000 to 800. Identified XGBoost as the best standalone model (ROC-AUC 0.97, PR-AUC 0.82) and Trimmed Mean aggregation as the best balanced ensemble.

Biomedical Abbreviation Detection and Expansion**Apr 2025**

BioBERT | DistilBERT | PLOD-CW-25 Dataset | Token Classification | Flask | Hugging Face | Focal Loss

- Built a biomedical NLP pipeline using token-level BIO tagging to detect abbreviations and their long-form expansions in scientific literature. Benchmarked BioBERT against DistilBERT on the PLOD-CW-25 dataset, with BioBERT achieving F1 of 0.8464 due to domain-specific pretraining on PubMed and PMC corpora. Experimented with Cross-Entropy, Weighted Cross-Entropy, and Focal Loss functions, with Focal Loss improving recall on minority token classes. Augmented training data using filtered PLODv2 subsets, deployed the best model to Hugging Face, and built a Flask-based chatbot interface for real-time abbreviation expansion with custom token-level logging for model monitoring.

Velour — Full-Stack E-Commerce Microservices Platform with AI Shopping Assistant

May 2025

Python | FastAPI | MongoDB | Motor | React | React Router | Axios | Docker | Nginx | JWT | GLM-4.5V | pytest

- **Backend and infrastructure:** Designed and built a 5-service microservices backend (Auth, Products, Cart, Search, AI) in FastAPI and MongoDB with the Motor async driver, containerised through Docker Compose and fronted by an Nginx reverse-proxy gateway that routes all frontend traffic to backend services, removing the need for a separate API gateway. Implemented atomic stock management using MongoDB's \$inc operator with conditional filters to prevent overselling under concurrent purchases, verified through a multi-threaded ThreadPoolExecutor load test, and secured services with JWT-based stateless authentication. Validated the system end-to-end with a pytest integration suite covering gateway routing, JWT auth, stock atomicity, cart lifecycle, checkout, and a full purchase flow, all running against the live Docker stack.
- **Frontend, search, and AI assistant:** Built a responsive React single-page application with React Router, Context API state management (Auth, Cart, Search), and protected routes, served through Nginx with SPA fallback routing and connected to the backend via Axios. Engineered a multi-signal product search engine combining keyword scoring, category aliases, natural-language price-ceiling parsing, and quality boosts (rating, popularity, discount), with an optional HuggingFace semantic-search fallback using all-MiniLM-L6-v2 sentence embeddings. Integrated GLM-4.5V as a structured-output shopping assistant exposed through a floating in-app chat widget (Vera) with conversation history, inline product cards, and live product-page context detection, injecting the 685-product catalogue and the user's order history into the model context window on every request to ground recommendations in real inventory and prior behaviour.

LLM Trading Insights System (Llama-2) | BSc Final Year Project, Trade for Sight (MoU)

Aug 2023 – Mar 2024

PyTorch | Llama-2 | LangChain | LlamaIndex | Pinecone | RAG | Chain-of-Thought Prompting

- Built under a formal MoU with Trade for Sight to solve a real financial domain problem: traders were unable to efficiently interpret complex graph-based market data, leading to slower decisions and declining platform engagement. Engineered an ETL pipeline transforming SQL-stored numerical trading data into structured natural language representations, enabling Llama-2 to perform domain-specific reasoning. Implemented Chain-of-Thought prompting to guide step-by-step decomposition of multi-variable graph data, and grounded all outputs using Pinecone vector search with LlamaIndex retrieval to prevent hallucinations. Outcomes: 30% increase in user engagement, 50% reduction in graph interpretation time, and 70% adoption among active traders within three months.

Llama-2 Augmented Mental Health Support System

Dec 2023

PyTorch | Llama-2 (7B) | QLoRA | PeFT | Gradio | Kaggle GPU P100 | Hugging Face

- Fine-tuned Llama-2 (7B) on the Amod mental health counselling conversations dataset to develop a virtual therapist capable of delivering empathetic, contextually appropriate responses. Applied QLoRA to quantise the model from 32-bit to 4-bit integer format, reducing GPU memory requirements from 28GB to 6GB and making fine-tuning feasible on a single Kaggle P100 GPU. Used PeFT with LoRA adapters to update less than 1% of model parameters, achieving domain adaptation at a fraction of full fine-tuning cost. Published the trained model to Hugging Face under Lohit20/Depressed_Llama-2-7b.

E-Commerce Visual Search System (CLIP + ViT)

Oct – Nov 2023

PyTorch | CLIP | Vision Transformer (ViT) | Pinecone | Streamlit | Colab GPU T4

- Built a multi-modal product search system allowing users to find products using either an image upload or a natural language description, eliminating reliance on precise keyword queries. Combined OpenAI CLIP with a Vision Transformer encoder to generate unified image-text embeddings, stored product embeddings in Pinecone, and retrieved the closest matches using cosine similarity. Enabled dual search modes within a single unified retrieval pipeline and deployed via a real-time Streamlit interface with results ranked by semantic similarity score.

Hybrid Movie Recommendation System (Multi-Armed Bandits + Content-Based Filtering)

Mar 2023

Python | Scikit-learn | Reinforcement Learning | Collaborative Filtering

- Designed a hybrid recommendation engine combining collaborative multi-armed bandits for adaptive exploration-based preference learning with content-based filtering for item-feature-driven recommendations using genre, cast, and metadata. The bandit component dynamically balanced exploration and exploitation to discover new user interests, directly addressing the cold-start problem. Evaluated recommendation diversity and accuracy metrics to validate that the hybrid approach outperformed either method independently.

Stock Price Prediction Using LSTM

Dec 2022

Python | Selenium | LSTM (PyTorch) | Tableau | NASDAQ

- Collected real-time stock price time-series data from NASDAQ using a Selenium-based web scraper, automating data ingestion across multiple tickers and trading sessions. Trained a multi-layer LSTM network to capture long-range temporal dependencies, achieving R^2 of 0.97 and MSE of 0.0680 on the held-out test set. Built an interactive Tableau dashboard presenting historical price trends, detected patterns, and 30-day forward projections for non-technical stakeholders.

Twitter Word Clustering (ETL + ML Pipeline)

Jun 2023

Apache Kafka | ETL | Scikit-learn | Python | Streaming Data

- Built a real-time data engineering pipeline using Apache Kafka to ingest, filter, and preprocess a continuous stream of Twitter data, handling schema variability and high-throughput ingestion reliably. Applied K-Means clustering with TF-IDF embeddings to group semantically similar tweets into coherent topic clusters, making the raw stream actionable for downstream sentiment analysis and trend monitoring.

Contextualised Course and Role Assignment (Ontology + NLP)

Dec 2022

Python | XML | SPARQL | RDFS | spaCy NLP

- Built a knowledge representation system that automatically assigned university courses to instructors by parsing teacher resumes with NLP entity extraction and skill classification, then matching extracted competencies against structured course requirement ontologies. Designed OWL/RDFS ontologies capturing qualification hierarchies and course prerequisites, and wrote SPARQL queries to perform logical inference and retrieve optimal course-instructor pairings via knowledge graph reasoning.

EDUCATION

- **MSc Data Science, Distinction** | University of Surrey, Guildford, UK | **Feb 2025 – Feb 2026**. *Modules:* Machine Learning and Data Mining, Data Science Principles and Practices, Natural Language Processing, Cloud Computing.
- **BSc Artificial Intelligence, CGPA 3.33, Three-Time Dean's List** | NUCES FAST University, Karachi, Pakistan | **Sep 2020 – Jun 2024**. *Modules:* Data Structures, Algorithms, Generative AI, Computer Vision, ML, NLP, Databases, OOP.
- **A-Levels** | Cedar College, Karachi, Pakistan | **Aug 2018 – Jun 2020**. *Subjects:* Computer Science, Mathematics, Physics, Chemistry.

CERTIFICATIONS

Orchestrating Workflows for GenAI Applications, DeepLearning.AI (**Jul 2025**) | Introduction to LLMs in Python, DataCamp (**Apr 2024**) | Generative AI Concepts, DataCamp (**Mar 2024**) | TensorFlow for AI, ML and Deep Learning, DeepLearning.AI (**Aug 2023**) | Ensemble Methods in Python, DataCamp (**Nov 2022**)

HONOURS AND AWARDS

- MSc Dissertation, Distinction, University of Surrey (**Nov 2025**)
- Three-Time Dean's List Honouree, NUCES FAST University (**Spring 2021, Spring 2023, Spring 2024**)
- Data Science PROCOM Competition, 7th place out of 30+ teams (**Spring 2023**)
- IEEE AI-Enabled TechEx Poster Competition, Finalist (**Fall 2022**)